Software Requirements Specification (SRS)

Nyaya Sahayak - AI-Based Legal Intelligence System

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Document Control

Version	Date	Author	Changes
1.0	January 2026	Development Team	Initial draft
2.0	January 28, 2026	Development Team	Updated tech stack, added lawyers script

Development Team:

- Muneeb Sawez
- Mohammed Fawaz Mulla
- Sayyad Abrar Ali
- Kaushik Rajesh Parvate

Client: New United Consultancy (Paralegal Firm)

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document provides a complete description of the functional and non-functional requirements for Nyaya Sahayak, an AI-powered legal intelligence platform developed for New United Consultancy. This document is intended for:

- Development team members
- Project stakeholders
- System administrators
- Future maintainers

1.2 Scope

Product Name: Nyaya Sahayak (Judicial Assistant)

Product Overview:

Nyaya Sahayak is a comprehensive web-based legal intelligence system that combines artificial intelligence, machine learning, and legal domain expertise to assist lawyers and legal professionals in:

- Case research and analysis
- Document management
- Outcome prediction
- Legal section identification
- Court presentation preparation
- Professional networking

System Components:

Internal Legal Intelligence System

- Deployed on law firm's local server
- Handles confidential case data
- AI-powered predictions and analysis
- Document repository
- Lawyers court script generation
- Counter questions preparation

Professional Networking Platform (Future Scope)

- Cloud-based Django application
- Connects lawyers with law students
- Job and internship marketplace
- Profile management

Primary Focus Areas:

- Banking and financial cases
- Criminal cases
- Civil and property disputes
- Cybercrime cases

What the System Will Do:

- Upload and analyze legal documents (FIRs, plaints, judgments)
- Extract structured information using NLP
- Find similar past cases using semantic search
- Predict case outcomes (conviction, bail, sentencing)
- Suggest relevant legal sections
- Generate court presentation scripts for lawyers
- Provide counter question preparation
- Provide AI chatbot for legal queries
- Maintain complete audit trail
- Facilitate lawyer-student networking

What the System Will NOT Do:

- Replace professional legal judgment
- Provide legally binding advice
- Make autonomous legal decisions
- Handle client-lawyer privileged communications
- Generate legally enforceable documents without lawyer review

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
AI	Artificial Intelligence
API	Application Programming Interface
BNS	Bharatiya Nyaya Sanhita (New Indian Penal Code)
CORS	Cross-Origin Resource Sharing
CrPC	Code of Criminal Procedure
CSRF	Cross-Site Request Forgery
FIR	First Information Report
IPC	Indian Penal Code
JWT	JSON Web Token
ML	Machine Learning
NER	Named Entity Recognition
NLP	Natural Language Processing
OCR	Optical Character Recognition

ORM	Object-Relational Mapping
RAG	Retrieval Augmented Generation
RBAC	Role-Based Access Control
REST	Representational State Transfer
SARFAESI	Securitisation and Reconstruction of Financial Assets and Enforcement of Security Interest Act
SHAP	SHapley Additive exPlanations
SRS	Software Requirements Specification
SSL/TLS	Secure Sockets Layer / Transport Layer Security
UI/UX	User Interface / User Experience

1.4 References

- Kaggle Dataset: Indian Supreme Court Judgments
    - URL: https://www.kaggle.com/datasets/vangap/indian-supreme-court-judgments
    - Contains: 34,000+ Supreme Court judgments

Django Documentation: https://docs.djangoproject.com/
React Documentation: https://react.dev/
Pinecone Vector Database: https://docs.pinecone.io/
Sentence Transformers: https://www.sbert.net/

1.5 Overview

This SRS document is organized as follows:

- **Section 2:** Provides overall system description, user characteristics, constraints
- **Section 3:** Details all functional requirements organized by feature
- **Section 4:** Specifies external interfaces (user, hardware, software)
- **Section 5:** Lists system requirements and technology stack
- **Section 6:** Describes non-functional requirements (performance, security)
- **Section 7:** Defines data requirements and sources
- **Section 8:** Contains appendices and supplementary information

2. Overall Description

2.1 Product Perspective

Nyaya Sahayak is a new, self-contained system and a one-stop solution, designed specifically for legal professionals. It consists of two major components:

- A: Core Implementation
- B: Future Scope

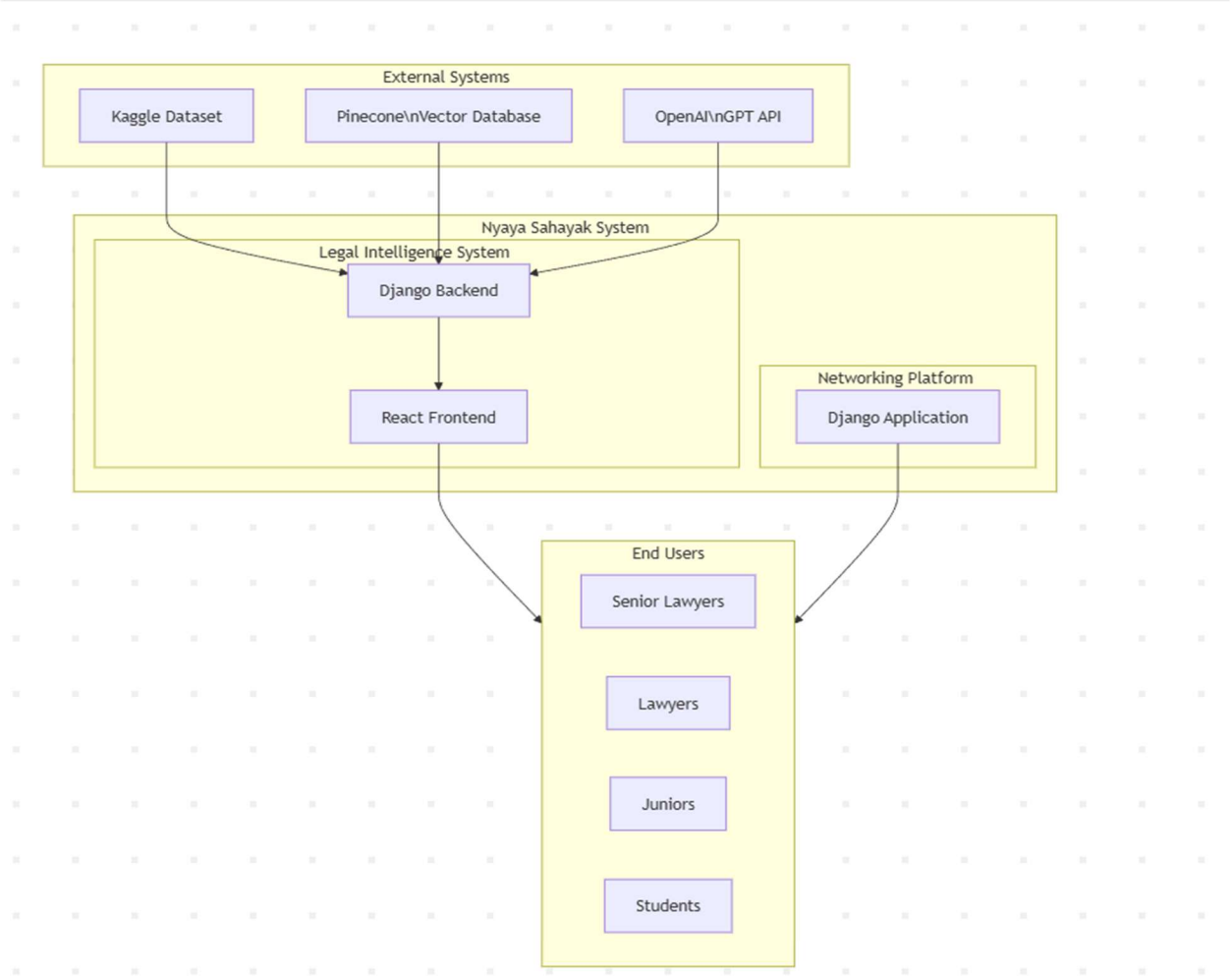
Component A: Internal Legal Intelligence System

- Standalone web application
- Deployed on law firm's infrastructure
- Operates independently but can be extended
- Integrates with existing file systems
- Does not replace but augments existing legal research tools

Component B: Professional Networking Platform

- Cloud-based web application
- Independent from Component A
- Multi-tenant architecture
- Can serve multiple law firms

System Context Diagram:



2.2 Product Functions

The system shall provide the following major functions:

F1: Document Management

- Upload legal documents (PDF, DOCX, images)
- OCR for scanned documents
- Version control
- Secure storage

F2: Information Extraction

- Automatic entity extraction (names, dates, sections)
- Case categorization
- Metadata generation
- Text summarization

F3: AI-Powered Search

- Semantic similarity search
- Find related case law
- Filter by multiple criteria
- Relevance ranking

F4: Predictive Analytics

- Conviction probability prediction
- Bail likelihood prediction
- Sentence duration estimation
- Legal section suggestion

F5: AI Legal Assistant

- Natural language Q&A chatbot
- Case law references
- Legal procedure guidance
- Context-aware responses

F6: Court Script Generation

- Generate structured court presentation scripts
- Include key arguments and precedents
- Customize by case type
- Export in multiple formats

F7: Counter Question Response

- Generate counter questions
- Possible objection
- Provide appropriate response for both.

F8: Knowledge Repository

- Centralized case database
- Searchable archive
- Tagging and categorization
- Access control of documents

F9: Explainable AI

- Show prediction reasoning
- Display influential factors
- Transparent decision-making
- Audit trail

F10: Professional Networking

- Lawyer and student profiles
- Job posting and applications
- Domain-based matching
- Communication tools

2.3 User Classes and Characteristics

2.3.1 User Class 1: Senior Lawyer / Administrator

Description:

- Experienced legal professionals
- Firm partners or senior associates
- High technical literacy

Responsibilities:

- Case oversight and approval
- Team management
- Report review and authorization
- Prediction validation

Permissions:

- Full system access

- User management
- Approve AI predictions
- Export all reports
- View audit logs
- Modify system settings
- Priority case marking

Technical Skills: Medium to High

Frequency of Use: Daily

Critical User: Yes

2.3.2 User Class 2: Lawyer

Description:

- Licensed legal practitioners
- Primary system users
- Moderate to high technical literacy

Responsibilities:

- Case research and analysis
- Document preparation
- Client representation
- Using AI tools for case strategy
- Court script preparation
- Counter question & objections preparation

Permissions:

- Upload and manage own cases
- Use all AI features
- Search and research
- Export reports
- Collaborate with team
- Generate court scripts & counter question

Technical Skills: Medium

Frequency of Use: Daily

Critical User: Yes

2.3.3 User Class 3: Junior Lawyer / Legal Associate

Description:

- Recently qualified lawyers
- Learning and supporting role
- Variable technical literacy

Responsibilities:

- Legal research
- Document drafting
- Case file preparation
- Assisting lawyers

Permissions:

- Research access
- View non-sensitive cases
- Use AI tools (with supervision)
- Limited export capabilities
- Requires approval for certain actions

Technical Skills: Low to Medium

Frequency of Use: Daily

Critical User: Moderate

2.3.4 User Class 4: Intern

Description:

- Law students or recent graduates
- No formal legal experience
- Basic legal knowledge
- Low technical literacy

Responsibilities:

- Basic research tasks
- Document organization
- Learning legal procedures

Permissions:

- Read-only access
- View public cases only

- Use chatbot for learning
- No export rights
- All actions require approval

Technical Skills: Low

Frequency of Use: Weekly

Critical User: No

2.3.5 User Class 5: Law Associates (Networking Platform)

Description:

- Law Professional
- Seeking internships or jobs
- Lawyer Connection
- Variable technical literacy

Responsibilities:

- Profile creation
- Job & intern applications
- Networking with lawyers

Permissions:

- Create and manage profile
- Browse job listings
- Apply to positions
- Message potential employers
- Connect & Chat

Technical Skills: Low to Medium

Frequency of Use: Weekly

Critical User: No

2.4 Operating Environment

OE-1: Client Operating Systems

- Windows 10/11
- macOS 11 or later
- Linux (Ubuntu 20.04 or later)

OE-2: Web Browsers

- Google Chrome 90+
- Mozilla Firefox 88+
- Microsoft Edge 90+
- Safari 14+ (macOS only)
- Minimum screen resolution: 1366 x 768

OE-3: Server Environment (Component A - Local)

- Operating System: Ubuntu Server 22.04 LTS
- Python: 3.11+
- Node.js: 18+ LTS
- PostgreSQL: 15+
- MongoDB: 7+
- Redis: 7+
- Nginx: 1.24+
- Docker: 24+

OE-4: Cloud Environment (Component B - Networking)

- Platform: Render
- Database: Managed PostgreSQL

OE-5: Network Requirements

- Internal Network: Gigabit Ethernet minimum
- Internet Connection: 10 Mbps minimum
- Firewall: Configured to allow HTTPS traffic
- VPN: Optional for remote access

OE-6: Hardware Requirements (Local Server)

- CPU: 8+ cores (Intel Xeon or AMD EPYC)
- RAM: 4 GB minimum, 16 GB recommended
- Storage: 256 GB SSD minimum, 1 TB recommended
- GPU: Optional (NVIDIA RTX 3060 12GB for faster ML inference)
- Network: Gigabit Ethernet port
- Backup: External storage or NAS for daily backups

2.5 Design and Implementation Constraints

C-1: Technology Stack Constraints

- Frontend MUST use JavaScript (React framework)
- Backend MUST use Django (Python framework)
- Cannot use proprietary software requiring licensing fees

- Must use open-source technologies where possible

C-2: Data Privacy Constraints

- All case data must remain on local server (Component A)
- Compliance with legal confidentiality requirements
- Personal data must be encrypted at rest and in transit

C-3: Legal and Regulatory Constraints

- System must not provide legally binding advice
- All AI predictions must be labeled as "decision support only"
- Lawyer verification required before any prediction is acted upon
- Must maintain audit logs for 1 years minimum

C-4: Performance Constraints

- API responses must return within 60 seconds (95th percentile)
- Document upload must complete within 120 seconds for 10MB files
- Similarity search must return results within 30 seconds
- System must handle 50 concurrent users

C-5: Security Constraints

- All communications must use HTTPS/TLS 1.3
- Passwords must be hashed with bcrypt (minimum 12 rounds)
- JWT tokens must expire within 24 hours
- Regular security audits mandatory

C-6: Development Constraints

- System developed free of cost for academic purposes
- Development team retains source code ownership
- 16-week development timeline
- Minimum 4 developers
- No external contractors

C-7: Infrastructure Constraints

- Local server deployment required for Component A
- Limited to single physical server initially
- Database backups required daily

C-8: Integration Constraints

- Must integrate with Kaggle dataset as primary data source

- Must use Pinecone or similar vector database
- Must use OpenAI API or compatible LLM service
- Limited API call budget for paid services

2.6 Assumptions and Dependencies

Assumptions:

A-1: Users have stable internet connectivity for cloud features

A-2: Law firm will provide dedicated server infrastructure

A-3: Users have basic computer literacy

A-4: Kaggle dataset remains available and accessible

A-5: OpenAI API or alternative LLM service remains operational

A-6: Legal regulations allow use of AI for decision support

A-7: Users will verify all AI predictions before taking legal action

A-8: Law firm has IT support for basic server maintenance

Dependencies:

D-1: External Data Source

- Kaggle Indian Supreme Court Judgments dataset
- 34,000+ court judgments
- CSV format availability

D-2: Third-Party Services

- Pinecone vector database (cloud service)
- OpenAI GPT-4 API (or alternative: Llama 3)
- Email service provider (for networking platform)

D-3: Open Source Libraries

- Django framework and ecosystem
- React and related packages
- PyTorch for ML models
- Sentence Transformers for embeddings
- PostgreSQL database
- MongoDB database
- Redis

D-4: Infrastructure

- Local server provisioned by law firm

D-5: Development Tools

- Git version control
- GitHub for repository hosting
- Docker for containerization
- Development IDEs (VS Code)

3. System Features and Requirements

3.1 Feature 1: User Authentication and Authorization

Priority: High

Risk: High

3.1.1 Description

The system shall provide secure user authentication and role-based access control to ensure only authorized users can access specific features and data.

3.1.2 Functional Requirements

FR-1.1: The system shall provide a login page where users enter email and password.

FR-1.2: The system shall hash all passwords using bcrypt with a minimum of 12 rounds before storing in database.

FR-1.3: The system shall generate a JWT token upon successful authentication, valid for 24 hours.

FR-1.4: The system shall support four user roles:

- Admin/Senior Lawyer
- Lawyer
- Junior Lawyer
- Intern

FR-1.5: The system shall enforce role-based permissions:

- Admins can access all features
- Lawyers can access research and prediction features
- Juniors require approval for sensitive operations
- Interns have read-only access

FR-1.6: The system shall provide password reset functionality via email verification.

FR-1.7: The system shall log all login attempts (successful and failed) with timestamp and IP address.

FR-1.11: The system shall allow admins to create, modify, and deactivate user accounts.

FR-1.12: The system shall enforce password complexity requirements:

- Minimum 8 characters
- At least one uppercase letter
- At least one lowercase letter
- At least one number
- At least one special character

3.2 Feature 2: Document Upload and Management

Priority: High

Risk: Medium

3.2.1 Description

The system shall allow users to upload legal documents in various formats, extract text content, and store them securely.

3.2.2 Functional Requirements

FR-2.1: The system shall accept document uploads in the following formats:

- PDF (.pdf)
- Microsoft Word (.docx)
- Images (.jpg, .jpeg, .png)

FR-2.2: The system shall enforce a maximum file size limit of 25 MB per upload.

FR-2.3: The system shall extract text from PDF and DOCX files automatically.

FR-2.4: The system shall perform OCR (Optical Character Recognition) on uploaded images to extract text.

FR-2.5: The system shall display upload progress indicator to users.

FR-2.6 The system shall store original files in MongoDB GridFS.

FR-2.7: The system shall store extracted text and metadata in PostgreSQL database.

FR-2.8: The system shall allow users to categorize documents as:

- FIR (First Information Report)
- Plaint
- Complaint

- Judgment
- Other

FR-2.9: The system shall automatically generate a unique document ID for each upload.

FR-2.10: The system shall track document version history when documents are updated.

FR-2.11: The system shall allow users to download original uploaded documents.

FR-2.12: The system shall display document metadata including:

- Upload date
- Uploaded by (user)
- File size
- Document type
- Processing status

FR-2.13: The system shall notify users upon successful upload and processing completion.

3.3 Feature 3: Case Information Extraction (NER)

Priority: High

Risk: Medium

3.3.1 Description

The system shall automatically extract structured information from legal documents using Named Entity Recognition and natural language processing techniques.

3.3.2 Functional Requirements

FR-3.1: The system shall extract the following entities from uploaded documents:

- Person names (petitioner, respondent, judges, lawyers)
- Organization names (courts, police stations, companies)
- Locations (cities, districts, states)
- Dates (filing date, judgment date, hearing dates)
- Legal sections (IPC, BNS, CrPC, special acts)
- Case numbers

FR-3.2: The system shall use pre-trained NLP models (spaCy or Legal-BERT) for entity extraction.

FR-3.3: The system shall achieve minimum 85% accuracy in entity extraction.

FR-3.4: The system shall categorize each case into one of the following types:

- Criminal
- Banking/Financial

- Civil
- Cybercrime
- Other

FR-3.5: The system shall extract case outcome from judgment text:

- Allowed
- Dismissed
- Quashed
- Convicted
- Acquitted

FR-3.6: The system shall generate a brief case summary (200-300 words) automatically.

FR-3.7: The system shall identify whether bail decision is present and extract:

- Bail granted
- Bail denied

FR-3.8: The system shall extract sentence duration for convicted cases (in months/years).

FR-3.9: The system shall store all extracted entities in PostgreSQL database.

FR-3.10: The system shall allow users to manually correct extracted information.

FR-3.11: The system shall display confidence scores for each extracted entity.

FR-3.12: The system shall flag low-confidence extractions (< 70%) for manual review.

3.4 Feature 4: Case Similarity Search

Priority: High

Risk: Medium

3.4.1 Description

The system shall enable users to find similar past cases using semantic search based on case facts and legal issues.

3.4.2 Functional Requirements

FR-4.1: The system shall generate embeddings (384-dimensional vectors) for all case documents using sentence-transformers.

FR-4.2: The system shall store embeddings in Pinecone vector database.

FR-4.3: The system shall accept natural language queries or case facts as search input.

FR-4.4: The system shall return top 10 most similar cases by default.

FR-4.5: The system shall allow users to adjust number of results (5-50 cases).

FR-4.6: The system shall display similarity score (0-100%) for each result.

FR-4.7: The system shall support filtering similar cases by:

- Case type (criminal, banking, civil, etc.)
- Date range
- Specific legal sections
- Court level
- Outcome type

FR-4.8: The system shall rank results by relevance (similarity score).

FR-4.9: The system shall display key excerpts from similar cases highlighting matching content.

FR-4.10: The system shall provide "More like this" functionality to find cases similar to a selected result.

FR-4.11: The system shall return similarity search results within 30 seconds.

FR-4.12: The system shall allow users to save search queries for future use.

FR-4.13: The system shall log all similarity searches in audit trail.

3.5 Feature 5: Legal Section Suggestion

Priority: High

Risk: High

3.5.1 Description

The system shall suggest relevant IPC/BNS sections based on case facts using a trained machine learning model.

3.5.2 Functional Requirements

FR-5.1: The system shall accept case facts as text input (minimum 100 words).

FR-5.2: The system shall use a fine-tuned BERT model for multi-label classification.

FR-5.3: The system shall predict top 5 relevant legal sections with confidence scores.

FR-5.4: The system shall support both IPC and BNS section identification.

FR-5.5: The system shall display sections in order of relevance (highest confidence first).

FR-5.6: The system shall provide brief description of each suggested section.

FR-5.7: The system shall allow users to accept or reject suggested sections.

FR-5.8: The system shall achieve minimum F1-score of 0.85 on test data.

FR-5.9: The system shall return suggestions within 30 seconds.

FR-5.10: The system shall display warning: "AI suggestion - requires lawyer verification".

FR-5.11: The system shall log all section suggestions and user actions (accept/reject).

3.6 Feature 6: Outcome Prediction Models

Priority: High

Risk: High

3.6.1 Description

The system shall predict case outcomes using machine learning models trained on historical data.

3.6.2 Functional Requirements - Conviction Prediction

FR-6.1: The system shall predict probability of conviction (0-100%) based on case features.

FR-6.2: The system shall use XGBoost classifier trained on minimum 5,000 cases.

FR-6.3: The system shall consider the following features:

- Legal sections applied
- Case type
- Number of sections
- Court tier
- Case duration
- Evidence indicators
- Compromise status

FR-6.4: The system shall achieve minimum ROC-AUC score of 0.80 on test data.

FR-6.5: The system shall display binary prediction: "Likely Convicted" or "Likely Acquitted".

FR-6.6: The system shall return prediction within 60 second.

3.6.3 Functional Requirements - Bail Prediction

FR-6.7: The system shall predict probability of bail being granted (0-100%).

FR-6.8: The system shall use Random Forest or XGBoost classifier.

FR-6.9: The system shall consider offense severity, sections, and flight risk factors.

FR-6.10: The system shall achieve minimum accuracy of 75% on test data.

FR-6.11: The system shall display binary prediction: "Bail Likely" or "Bail Unlikely".

3.6.4 Functional Requirements - Sentence Prediction

FR-6.12: The system shall predict sentence duration in months for convicted cases.

FR-6.13: The system shall use XGBoost regressor.

FR-6.14: The system shall provide range estimate (minimum-maximum months).

FR-6.15: The system shall achieve Mean Absolute Error (MAE) of less than 6 months.

3.6.5 Common Prediction Requirements

FR-6.16: All predictions shall be clearly labeled as "Decision Support Only - Not Legal Advice".

FR-6.17: The system shall require lawyer approval before predictions are stored permanently.

FR-6.18: The system shall display warning message before showing predictions.

FR-6.19: The system shall log all prediction requests with user ID, timestamp, and case ID.

3.7 Feature 7: AI Legal Chatbot (RAG)

Priority: Medium

Risk: Medium

3.7.1 Description

The system shall provide an AI-powered chatbot that answers legal queries using Retrieval Augmented Generation (RAG) with case law context.

3.7.2 Functional Requirements

FR-7.1: The system shall provide a chat interface accessible from main navigation.

FR-7.2: The system shall accept natural language questions in English.

FR-7.3: The system shall use RAG architecture:

- Retrieve relevant case documents from vector database
- Provide context to Large Language Model
- Generate contextual answers

FR-7.4: The system shall use OpenAI GPT-4 API or self-hosted Llama 3 as LLM.

FR-7.5: The system shall retrieve top 5 most relevant case excerpts for each query.

FR-7.6: The system shall cite source cases in responses with case numbers.

FR-7.7: The system shall provide clickable links to cited cases.

FR-7.8: The system shall return responses within 60 seconds.

FR-7.9: The system shall maintain conversation history within a session.

FR-7.10: The system shall allow users to clear chat history.

FR-7.11: The system shall limit conversation to legal topics only.

FR-7.12: The system shall display disclaimer: "AI-generated response - verify with legal professional".

FR-7.13: The system shall handle up to 10 conversation turns per session.

FR-7.14: The system shall log all chatbot interactions for audit purposes.

FR-7.15: The system shall not store confidential client information in chat logs.

FR-7.16: The system shall provide suggested follow-up questions.

3.8 Feature 8: Lawyers Court Script & Counter question Generation

Priority: High

Risk: Medium

3.8.1 Description

The system shall generate structured court presentation scripts for lawyers to use during hearings, including key arguments, precedents, and procedural steps.

3.8.2 Functional Requirements

FR-8.1: The system shall provide a "Generate Court Script" option for each case.

FR-8.2: The system shall accept the following inputs from lawyer:

- Case facts summary
- Key arguments to emphasize
- Desired outcome (relief sought)
- Hearing type (bail hearing, final arguments, etc.)
- Court level (District, High Court, Supreme Court)

FR-8.3: The system shall generate a structured script containing:

- Opening statement
- Statement of facts
- Legal arguments with section references
- Relevant precedent citations
- Counter-arguments and rebuttals
- Closing submissions
- Prayer for relief

FR-8.4: The system shall retrieve and include 3-5 relevant precedent cases from database.

FR-8.5: The system shall format precedents with:

- Case name and citation
- Key legal principle
- How it applies to current case
- Quoted relevant portion (2-3 sentences maximum)

FR-8.6: The system shall organize script in logical flow:

Introduction and greetings
Case background
Points of law
Supporting case law
Conclusion and prayer

FR-8.7: The system shall adapt language formality based on court level:

- District Court: Standard formal language
- High Court: More formal with honorifics
- Supreme Court: Highly formal with traditional language

FR-8.8: The system shall include procedural reminders:

- Documents to be filed
- Time limits to mention
- Required formalities

FR-8.9: The system shall generate scripts in the following formats:

- PDF (formatted for printing)
- DOCX (editable in Microsoft Word)
- HTML (viewable in browser)

FR-8.10: The system shall allow lawyers to edit generated scripts before saving.

FR-8.11: The system shall save script history for each case (version control).

FR-8.12: The system shall include timestamps in generated scripts.

FR-8.13: The system shall generate scripts within 130 seconds.

FR-8.14: The system shall highlight critical sections for easy reference during hearing.

FR-8.15: The system shall include a checklist section for pre-hearing preparation.

FR-8.16: The system shall add disclaimer footer: "AI-generated script - review and customize before use".

FR-8.17: The system shall allow lawyers to specify tone:

- Aggressive (prosecution/plaintiff)
- Defensive (defending accused)
- Neutral (factual presentation)

FR-8.18: The system shall track script usage:

- Generated count
- Downloaded count
- Edited count

FR-8.19: The system shall allow sharing scripts with colleagues (within firm).

FR-8.20: The system shall log all script generation activities in audit trail.

3.9 Feature 9: Case Repository and Search

Priority: High

Risk: Low

3.9.1 Description

The system shall maintain a centralized repository of all cases with comprehensive search and filtering capabilities.

3.9.2 Functional Requirements

FR-9.1: The system shall display all cases in a paginated list (25 cases per page).

FR-9.2: The system shall provide full-text search across all case documents.

FR-9.3: The system shall support advanced filtering by:

- Case type
- Court name
- Date range (filing date, judgment date)
- Legal sections
- Outcome
- Uploaded by user
- Tags

FR-9.4: The system shall allow users to sort cases by:

- Date (newest/oldest)
- Relevance (for search results)

- Case number
- Alphabetical (by title)

FR-9.5: The system shall provide case detail view showing:

- Case number and title
- Parties involved
- Court and judge(s)
- Legal sections
- Case type and category
- Key dates
- Extracted entities
- Document preview
- AI predictions (if generated)

FR-9.6: The system shall allow users to add tags to cases for organization.

FR-9.7: The system shall support creating custom collections/folders of cases.

FR-9.8: The system shall allow users to add notes to cases (private to user).

FR-9.9: The system shall highlight recently added cases (within last 7 days).

FR-9.10: The system shall provide "Quick Actions" for each case:

- View details
- Download document
- Generate predictions
- Find similar cases
- Generate court script
- Export report

FR-9.11: The system shall implement full-text search returning results within 2 seconds.

FR-9.12: The system shall save user search preferences.

3.10 Feature 10: Explainability and Audit System

Priority: Medium

Risk: Low

3.10.1 Description

The system shall provide transparent explanations for all AI decisions and maintain comprehensive audit logs of user activities.

3.10.2 Functional Requirements - Explainability

FR-10.1: The system shall generate SHAP (SHapley Additive exPlanations) values for all ML predictions.

FR-10.2: The system shall display top 5 factors influencing each prediction with impact percentages.

FR-10.3: The system shall categorize factors as:

- Positive factors (increasing probability)
- Negative factors (decreasing probability)
- Neutral factors (minimal impact)

FR-10.4: The system shall visualize feature importance using bar charts or waterfall plots.

FR-10.5: The system shall generate human-readable explanations in plain English:

- Example: "Conviction predicted because Section 420 applied (+25%), weak alibi evidence (+18%)"

FR-10.6: The system shall show base probability and how each factor adjusts it.

FR-10.7: The system shall allow users to export explanation reports as PDF.

3.10.3 Functional Requirements - Audit System

FR-10.8: The system shall log the following user actions:

- Login/logout
- Case upload
- Case view
- Case edit/delete
- Prediction request
- Script generation
- Report export
- User account changes
- System settings modifications

FR-10.9: Each audit log entry shall capture:

- User ID and name
- Action type
- Case ID (if applicable)
- Timestamp (date and time)
- IP address
- Browser/device information

- Action result (success/failure)

FR-10.10: The system shall provide audit log dashboard accessible to admins and senior lawyers.

FR-10.11: The system shall allow filtering audit logs by:

- User
- Action type
- Date range
- Case ID

FR-10.12: The system shall allow exporting filtered audit logs as CSV or PDF.

FR-10.13: The system shall retain audit logs for minimum 7 years.

FR-10.14: The system shall prevent modification or deletion of audit logs (append-only).

FR-10.15: The system shall send email alerts for suspicious activities:

- Multiple failed login attempts
- Bulk data exports
- Unauthorized access attempts
- Administrative changes

FR-10.16: The system shall display user activity summary:

- Total logins this month
- Cases accessed
- Predictions generated
- Reports exported

3.11 Feature 11: Report Generation and Export

Priority: Medium

Risk: Low

3.11.1 Description

The system shall allow users to generate and export comprehensive reports for cases.

3.11.2 Functional Requirements

FR-11.1: The system shall generate case analysis reports containing:

- Case summary
- Extracted information
- Similar cases found

- AI predictions (if generated)
- Court script (if generated)
- Timeline of events

FR-11.2: The system shall support export formats:

- PDF
- Microsoft Word (.docx)
- HTML

FR-11.3: The system shall allow users to customize report content:

- Select sections to include
- Choose level of detail
- Add custom notes
- Include/exclude AI predictions

FR-11.4: The system shall include firm branding in reports:

- Firm logo
- Firm name and address
- Report generation date
- Prepared by (lawyer name)

FR-11.5: The system shall generate watermark on reports: "Confidential - Internal Use Only".

FR-11.6: Junior lawyers and interns shall require senior lawyer approval before exporting reports.

FR-11.7: The system shall track all report exports in audit logs.

FR-11.8: The system shall allow bulk export of multiple cases.

FR-11.9: The system shall generate reports within 10 seconds.

FR-11.10: The system shall include table of contents for multi-page reports.

3.12 Feature 12: Professional Networking Platform (Future scope)

Priority: Medium

Risk: Low

3.12.1 Description

A separate cloud-based Django application where lawyers can post job/internship opportunities and law students can apply.

3.12.2 Functional Requirements - Lawyer Profiles

FR-12.1: Lawyers shall be able to create professional profiles with:

- Name and photo
- Firm name and location
- Years of experience
- Areas of specialization (criminal, banking, civil, etc.)
- Education and certifications
- Contact information

FR-12.2: Lawyers shall be able to post job openings with:

- Job title
- Job type (full-time, part-time, internship)
- Legal domain (criminal, banking, civil, etc.)
- Location
- Experience required
- Job description
- Application deadline

FR-12.3: Lawyers shall be able to manage posted jobs:

- Edit job details
- Mark as filled/closed
- Delete postings

FR-12.4: Lawyers shall be able to view applications with:

- Applicant profile
- Resume/CV
- Cover letter
- Application date

FR-12.5: Lawyers shall be able to change application status:

- Under review
- Shortlisted
- Rejected
- Accepted

3.12.3 Functional Requirements - Student Profiles

FR-12.6: Students shall be able to create profiles with:

- Name and photo

- Law school and year
- Areas of interest
- Education details
- Skills and achievements
- Resume upload

FR-12.7: Students shall be able to browse job listings with filters:

- Job type (internship, full-time)
- Legal domain
- Location
- Experience level

FR-12.8: Students shall be able to apply to positions by:

- Submitting cover letter
- Attaching resume
- Answering optional screening questions

FR-12.9: Students shall be able to track application status.

FR-12.10: Students shall receive email notifications when:

- Application is reviewed
- Status changes
- New matching jobs are posted

3.12.4 Functional Requirements - Matching and Communication

FR-12.11: The system shall recommend jobs to students based on:

- Areas of interest
- Location preferences
- Education level
- Skills

FR-12.12: The system shall recommend candidates to lawyers based on:

- Job requirements
- Student qualifications
- Domain match

FR-12.13: The system shall provide in-app messaging between lawyers and applicants.

FR-12.14: The system shall send email notifications for new messages.

FR-12.15: The system shall provide analytics dashboard for lawyers:

- Total applications received
- Applications by status
- Popular job postings

FR-12.16: The system shall support searching lawyers by:

- Location
- Specialization
- Firm name

FR-12.17: The system shall allow ratings and reviews (optional feature).

3.13 Feature 13: System Administration

Priority: High

Risk: Medium

3.13.1 Description

The system shall provide administrative capabilities for user management, system configuration, and monitoring.

3.13.2 Functional Requirements

FR-13.1: Admins shall be able to create new user accounts with:

- Name and email
- Role assignment
- Initial password (must be changed on first login)

FR-13.2: Admins shall be able to modify user details:

- Change role
- Reset password
- Update email
- Deactivate/activate account

FR-13.3: Admins shall be able to view system statistics:

- Total users (by role)
- Total cases
- Total predictions generated
- Storage usage
- API usage

FR-13.4: Admins shall be able to configure system settings:

- Maximum file upload size
- Session timeout duration
- Password complexity rules
- API rate limits

FR-13.5: Admins shall be able to view system logs:

- Application errors
- API errors
- Database connection issues

FR-13.6: Admins shall be able to schedule automated backups:

- Database backups
- Document backups
- Backup frequency
- Retention period

FR-13.7: Admins shall be able to restore from backups.

FR-13.8: Admins shall receive email alerts for:

- System errors
- Failed backups
- High resource usage
- Security incidents

FR-13.9: Admins shall be able to approve/deny junior user requests for:

- Sensitive case access
- Report exports
- Prediction approvals

FR-13.10: Admins shall be able to delete cases permanently (after soft delete period).

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4. External Interface Requirements

4.1 User Interfaces

4.1.1 General UI Requirements

UI-1: The user interface shall be web-based and responsive, working on desktop (1920x1080) down to tablet (768x1024) screen sizes.

UI-2: The UI shall use modern design principles with consistent color scheme matching law firm branding.

UI-3: The UI shall provide clear navigation with main menu always visible.

UI-4: The UI shall display loading indicators for operations taking more than 2 seconds.

UI-5: The UI shall show appropriate error messages with clear instructions when operations fail.

UI-6: The UI shall use icons with text labels for better accessibility.

UI-7: The UI shall support keyboard navigation for all features.

UI-8: The UI shall provide breadcrumb navigation showing user's current location.

UI-9: The UI shall display user's name and role in header.

UI-10: The UI shall provide help tooltips for complex features.

4.1.2 Specific UI Components

Login Page:

- Clean, professional design
- Email and password fields
- "Remember me" checkbox
- "Forgot password" link
- Firm logo prominently displayed

Dashboard:

- Summary statistics (total cases, pending approvals, recent activity)
- Quick action buttons (Upload Case, Search, New Prediction)
- Recent cases list
- Notifications panel

Case Upload Page:

- Drag-and-drop file upload area
- File type and size restrictions clearly displayed
- Upload progress bar
- Preview of uploaded document

Case Detail Page:

- Tabbed interface:
- Overview (metadata)
- Document viewer

- Predictions
- Similar cases
- Court script
- Timeline
- Action buttons in header (Edit, Delete, Export, Generate Script)

Search Results Page:

- Search bar at top
- Filters sidebar (left)
- Results grid/list (center)
- Pagination controls (bottom)

Chatbot Interface:

- Chat window overlaying main content OR dedicated page
- Message history
- Input box with send button
- Example questions
- Clear conversation button

Court Script Generator:

- Form with input fields (facts, arguments, hearing type)
- Generate button
- Preview pane
- Edit interface
- Export options

Admin Dashboard:

- User management table
- System statistics cards
- Audit log viewer
- Settings panel

4.1.3 UI Mockup References

Detailed UI mockups shall be created during design phase and approved by stakeholders before development.

4.2 Hardware Interfaces

HI-1: The system shall interface with local file system for:

- Storing uploaded documents
- Reading configuration files

- Writing log files

HI-2: The system shall interface with network hardware:

- Network interface cards for server communication
- No special network hardware required

HI-3: The system shall utilize GPU (if available) for:

- Faster ML model inference
- Parallel processing of embeddings
- Optional, not required

HI-4: The system shall not directly interface with any specialized hardware devices.

4.3 Software Interfaces

4.3.1 External API Interfaces

SI-1: Pinecone Vector Database API

- Purpose: Store and query vector embeddings
- Communication: HTTPS REST API
- Data format: JSON
- Authentication: API key
- Operations:
- Insert vectors
- Query similar vectors
- Update metadata
- Delete vectors

SI-2: OpenAI API (or Llama 3 API)

- Purpose: Large Language Model for chatbot
- Communication: HTTPS REST API
- Data format: JSON
- Authentication: API key (OpenAI) or None (self-hosted Llama)
- Operations:
- Chat completion
- Embeddings generation

SI-3: Email Service API (SendGrid or similar)

- Purpose: Send notification emails (networking platform)
- Communication: HTTPS REST API
- Data format: JSON

- Authentication: API key
- Operations:
- Send single email
- Send bulk emails

4.3.2 Database Interfaces

SI-4: PostgreSQL Database

- Purpose: Store structured data (users, cases, predictions)
- Communication: PostgreSQL wire protocol
- Port: 5432
- Authentication: Username/password
- ORM: Django ORM

SI-5: MongoDB Database

- Purpose: Store unstructured data (full documents, raw uploads)
- Communication: MongoDB wire protocol
- Port: 27017
- Authentication: Username/password
- Driver: PyMongo

SI-6: Redis Cache

- Purpose: Session storage and caching
- Communication: Redis protocol
- Port: 6379
- Authentication: Password (optional)
- Driver: django-redis

4.3.3 Operating System Interfaces

SI-7: The system shall interface with Ubuntu Linux OS for:

- File I/O operations
- Process management
- Network socket operations
- System logging

4.4 Communication Interfaces

CI-1: HTTP/HTTPS Protocol

- All client-server communication shall use HTTPS (TLS 1.3)
- Port 443 for production
- Port 3000 (frontend dev), 8000 (backend dev) for development

CI-2: WebSocket Protocol (Optional)

- Real-time notifications for users
- Chatbot streaming responses
- Port 443 (wss://)

CI-3: SMTP Protocol

- Email notifications
- Port 587 (TLS) or 465 (SSL)

CI-4: DNS

- Domain name resolution
- Port 53

CI-5: API Communication

- RESTful architecture
- JSON data format
- Standard HTTP methods (GET, POST, PUT, DELETE)
- JWT token in Authorization header

CI-6: Database Communication

- PostgreSQL: Port 5432
- MongoDB: Port 27017
- Redis: Port 6379
- All connections within local network or encrypted tunnel

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5. System Requirements

5.1 Technology Stack

5.1.1 Frontend Technologies

Component	Technology	Version	Purpose
UI Library	React	18.2+	Component-based user interface
Language	JavaScript (ES6+)	-	Primary programming language
Build Tool	Vite	4.0+	Fast build and development server
Styling	Tailwind CSS	3.3+	Utility-first CSS

			framework
HTTP Client	Axios	1.4+	API communication
State Management	Zustand	4.3+	Lightweight state management
Form Handling	React Hook Form	7.45+	Form validation and handling
Routing	React Router	6.14+	Client-side routing
Charts	Recharts	2.7+	Data visualization
Icons	Lucide React	0.263+	Icon library
PDF Viewer	react-pdf	7.3+	Display PDF documents
Date Handling	date-fns	2.30+	Date manipulation

5.1.2 Backend Technologies

Component	Technology	Version	Purpose
Framework	Django	5.0+	Web application framework
Language	Python	3.11+	Primary programming language
REST API	Django REST Framework	3.14+	RESTful API creation
Authentication	djangorestframework-simplejwt	5.3+	JWT authentication
CORS	django-cors-headers	4.2+	Cross-origin resource sharing
Environment	python-decouple	3.8+	Environment variable management
File Upload	django-storages	1.13+	File storage backends
Task Queue	Celery	5.3+	Asynchronous task processing
Message Broker	Redis	7.0+	Task queue messaging
Admin Interface	Django Admin	Built-in	Administrative interface

5.1.3 Machine Learning & NLP

Component	Technology	Version	Purpose
Deep Learning	PyTorch	2.0+	Neural network framework
Transformers	Hugging Face Transformers	4.30+	Pre-trained models (BERT)
Traditional ML	scikit-learn	1.3+	Classical ML algorithms
Gradient Boosting	XGBoost	2.0+	Tree-based models

NER	spaCy	3.6+	Named entity recognition
Embeddings	sentence-transformers	2.2+	Text vectorization
Explainability	SHAP	0.42+	Model interpretability
Data Processing	pandas	2.0+	Data manipulation
Numerical	NumPy	1.24+	Numerical computing

5.1.4 Databases & Storage

Component	Technology	Version	Purpose
Relational DB	PostgreSQL	15+	Structured data, metadata
Document DB	MongoDB	7.0+	Unstructured documents
Vector DB	Pinecone	Cloud	Embedding storage, similarity search
Cache	Redis	7.0+	Session, temporary data
ORM	Django ORM	Built-in	Database abstraction
MongoDB Driver	PyMongo	4.4+	MongoDB Python driver

5.1.5 DevOps & Deployment

Component	Technology	Version	Purpose
Containerization	Docker	24.0+	Application packaging
Orchestration	Docker Compose	2.20+	Multi-container management
Web Server	Nginx	1.24+	Reverse proxy, static files
WSGI Server	Gunicorn	21.0+	Python WSGI HTTP server
Process Manager	Supervisor	4.2+	Process control
SSL/TLS	Let's Encrypt	-	Free SSL certificates
Version Control	Git	2.40+	Source code management

5.1.6 Development Tools

Component	Technology	Purpose
Code Editor	VS Code	Primary IDE
API Testing	Postman / Thunder Client	API testing
Database Client	DBeaver / pgAdmin	PostgreSQL management
MongoDB Client	MongoDB Compass	MongoDB management

Python Env	venv / virtualenv	Virtual environment
Package Manager	npm	JavaScript packages
Package Manager	pip	Python packages

5.1.7 Third-Party Services

Service	Purpose	Cost Model
Pinecone	Vector database	Free tier + usage-based
OpenAI API	LLM for chatbot	Pay-per-token
SendGrid	Email notifications	Free tier + volume-based
AWS/Azure	Cloud hosting (networking)	Pay-as-you-go
Let's Encrypt	SSL certificates	Free

5.2 Development Environment Requirements

DE-1: Developers shall use Ubuntu 22.04 LTS or macOS 12+ or Windows 11 with WSL2.

DE-2: Minimum development machine specifications:

- CPU: Intel i5 / AMD Ryzen 5 or better
- RAM: 16 GB
- Storage: 50 GB free space (SSD recommended)

DE-3: Required software installations:

- Python 3.11+
- Node.js 18+ LTS
- PostgreSQL 15+
- MongoDB 7+
- Redis 7+
- Docker 24+
- Git 2.40+

DE-4: Recommended VS Code extensions:

- Python
- Pylance
- Django
- ESLint
- Prettier
- Tailwind CSS IntelliSense
- Docker

5.3 Production Environment Requirements

PE-1: Server Specifications (Local Deployment)

- Operating System: Ubuntu Server 22.04 LTS
- CPU: 8+ cores (Intel Xeon or AMD EPYC)
- RAM: 32 GB minimum, 64 GB recommended
- Storage: 500 GB SSD minimum, 1 TB recommended
- Network: Gigabit Ethernet
- Backup: External storage or NAS

PE-2: Network Requirements

- Internal network: Gigabit Ethernet minimum
- Internet: 50 Mbps symmetric minimum
- Firewall: Allow ports 80, 443 (HTTPS)
- VPN: Optional for remote access

PE-3: Backup Requirements

- Daily automated backups
- 30-day retention period
- Offsite backup storage
- Regular backup testing

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6. Non-Functional Requirements

6.1 Performance Requirements

P-1: The system shall handle 50 concurrent users without performance degradation.

P-2: API response time shall be less than 500ms for 95% of requests (p95).

P-3: Document upload shall complete within 60 seconds for files up to 25 MB.

P-4: Similarity search shall return results within 60 seconds.

P-5: ML model predictions shall complete within 30 seconds.

P-6: Chatbot responses shall be generated within 5 seconds.

P-7: Court script generation shall complete within 30 seconds.

P-8: Database queries shall execute within 100ms for 95% of queries.

P-9: Page load time shall be less than 2 seconds on broadband connection.

P-10: The system shall support file uploads up to 25 MB.

P-11: The system shall handle up to 1,000 cases in the repository initially, scalable to 10,000+.

P-12: The system shall support storing up to 100,000 document pages.

6.2 Safety Requirements

S-1: The system shall automatically backup all data daily to prevent data loss.

S-2: The system shall provide confirmation dialogs before deleting any data.

S-3: The system shall implement soft delete, retaining deleted data for 30 days before permanent deletion.

S-4: The system shall prevent data corruption by implementing database transactions.

S-5: The system shall validate all user inputs to prevent injection attacks.

S-6: The system shall limit API rate to prevent denial-of-service attacks.

S-7: The system shall gracefully handle failures without exposing sensitive information.

S-8: The system shall include disclaimer on all AI predictions: "This is AI-generated decision support. Verify with legal professional before taking action."

6.3 Security Requirements

SE-1: Authentication

- All users must authenticate with email and password
- Passwords must be hashed with bcrypt (12 rounds minimum)
- JWT tokens for session management (24-hour expiry)
- Account lockout after 5 failed login attempts
- Optional two-factor authentication for admins

SE-2: Authorization

- Role-based access control (RBAC)
- Principle of least privilege
- Resource-level permissions
- API endpoint protection

SE-3: Data Protection

- All data transmission via HTTPS/TLS 1.3
- Database encryption at rest
- Sensitive fields (passwords) never logged
- Personal data anonymization in logs

SE-4: Input Validation

- Server-side validation for all inputs

- Protection against SQL injection
- Protection against XSS attacks
- CSRF token validation

SE-5: API Security

- Rate limiting (100 requests/minute per user)
- API key authentication for external integrations
- CORS policy enforcement
- Request size limits

SE-6: File Upload Security

- File type validation (whitelist approach)
- Virus scanning before processing
- File size limits (25 MB)
- Storage outside web root

SE-7: Session Management

- Secure session cookies (HttpOnly, Secure flags)
- Session timeout after 30 minutes inactivity
- Session invalidation on logout
- Prevention of session fixation

SE-8: Logging and Monitoring

- Log all authentication attempts
- Log all data access and modifications
- Alert on suspicious activities
- Regular security audit log reviews

SE-9: Vulnerability Management

- Regular dependency updates
- Security patches applied within 7 days
- Quarterly security audits
- Penetration testing before production

SE-10: Compliance

- No storage of sensitive client data without encryption
- Compliance with data protection regulations
- Regular compliance audits

6.4 Software Quality Attributes

6.4.1 Reliability

R-1: The system shall have 99.5% uptime (excluding planned maintenance).

R-2: The system shall recover automatically from crashes within 5 minutes.

R-3: The system shall maintain data integrity during failures.

R-4: The system shall gracefully degrade if external services (OpenAI, Pinecone) are unavailable.

R-5: The system shall provide alternative functionality when AI services fail.

6.4.2 Availability

A-1: The system shall be available 24/7 except for planned maintenance.

A-2: Planned maintenance shall be scheduled during off-hours (midnight to 4 AM).

A-3: Users shall be notified 48 hours before planned maintenance.

A-4: Critical bugs shall be fixed within 24 hours.

A-5: Non-critical bugs shall be fixed within 7 days.

6.4.3 Maintainability

M-1: The code shall follow PEP 8 style guide (Python) and Airbnb style guide (JavaScript).

M-2: All functions shall have documentation comments.

M-3: The system shall have comprehensive unit tests (80% code coverage minimum).

M-4: The system shall have integration tests for critical workflows.

M-5: The codebase shall be modular with clear separation of concerns.

M-6: Configuration shall be externalized in environment files.

M-7: The system shall use dependency injection for testability.

M-8: Database migrations shall be versioned and reversible.

6.4.4 Scalability

SC-1: The system shall support horizontal scaling by adding more servers.

SC-2: The database shall support read replicas for scaling reads.

SC-3: The system shall use caching to reduce database load.

SC-4: The system shall support multi-firm deployment in future.

SC-5: The architecture shall allow adding new ML models without major changes.

6.4.5 Usability

U-1: New users shall be able to upload a case within 5 minutes of first login (after training).

U-2: The system shall provide context-sensitive help for all features.

U-3: Error messages shall be clear and actionable.

U-4: The system shall be accessible (WCAG 2.1 Level AA compliance target).

U-5: The system shall work on modern browsers without plugins.

U-6: The system shall provide keyboard shortcuts for power users.

U-7: The UI shall be intuitive, requiring minimal training.

6.4.6 Portability

PO-1: The system shall run on any Linux distribution with Docker support.

PO-2: The system shall use standard protocols (HTTP, SQL) for interoperability.

PO-3: The system shall export data in standard formats (CSV, JSON, PDF).

PO-4: The system shall allow migration to different cloud providers.

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7. Data Requirements

7.1 Primary Data Source

Source: Kaggle - Indian Supreme Court Judgments Dataset

URL: https://www.kaggle.com/datasets/vangap/indian-supreme-court-judgments

Dataset Details:

- **Total Judgments:** 34,000+ Supreme Court cases
- **Format:** CSV file
- **Size:** Approximately 500 MB
- **Coverage:** Diverse legal domains
- **Time Period:** Historical judgments up to recent years
- **Language:** English

Columns in Dataset:

Case Title
Case Number

Judge Name(s)
Petitioner
Respondent
Judgment Text (full)
Date of Judgment
Citation

Usage Plan:

- Download complete dataset from Kaggle
- Select 10,000 cases for initial system deployment:
- Banking/Financial: 2,500 cases
- Criminal: 4,000 cases
- Civil: 2,500 cases
- Others: 1,000 cases
- Remaining cases reserved for future expansion

7.2 Data Processing Requirements

DP-1: All data processing shall be done in Google Sheets preprocessing pipeline before import.

DP-2: Google Sheets pipeline shall perform:

- Data cleaning (remove duplicates, fix encoding)
- Text extraction and normalization
- Case categorization
- Feature engineering
- Quality validation

DP-3: Pipeline shall produce two outputs:

- **For RAG:** Vectorized embeddings uploaded to Pinecone
- **For ML:** JSON files (train/validation/test split)

DP-4: Data split for ML training:

- Training set: 70% (7,000 cases)
- Validation set: 15% (1,500 cases)
- Test set: 15% (1,500 cases)

DP-5: Embeddings shall be 384-dimensional vectors (sentence-transformers: all-MiniLM-L6-v2).

DP-6: All processed data shall be validated for:

- Completeness (no null critical fields)
- Accuracy (sample manual verification)

- Consistency (standardized formats)

7.3 Data Storage Requirements

DS-1: PostgreSQL Database

- Store structured metadata (users, cases, predictions)
- Total estimated size: 500 MB for 10,000 cases
- Daily incremental backups
- Weekly full backups

DS-2: MongoDB Database

- Store full judgment text
- Store original uploaded files (GridFS)
- Total estimated size: 5-10 GB
- Daily incremental backups

DS-3: Pinecone Vector Database

- Store 384-dimensional embeddings
- Estimated: 10,000 vectors × 384 dimensions
- Cloud-managed (no local backup needed)

DS-4: Redis Cache

- Session data
- Frequently accessed case metadata
- Estimated size: 100 MB
- Volatile storage (rebuilt on restart)

DS-5: File System

- Logs: 1 GB retention (30 days)
- Temporary files: 5 GB
- Backup storage: 20 GB

7.4 Data Retention and Archival

DR-1: Active case data shall be retained indefinitely.

DR-2: Deleted cases shall be soft-deleted and retained for 30 days before permanent deletion.

DR-3: Audit logs shall be retained for 7 years minimum (legal compliance).

DR-4: Temporary files shall be purged after 24 hours.

DR-5: Old backups shall be archived after 90 days (monthly backups retained for 1 year).

7.5 Data Quality Requirements

DQ-1: Case data completeness: 95% of fields populated.

DQ-2: Section extraction accuracy: 90% minimum.

DQ-3: Case categorization accuracy: 85% minimum.

DQ-4: Entity extraction accuracy (NER): 85% minimum.

DQ-5: Duplicate detection: 99% effective (no duplicate cases in system).

7.6 Data Privacy and Compliance

DC-1: All user passwords shall be hashed (never stored in plain text).

DC-2: Personal identifiable information (PII) shall be encrypted at rest.

DC-3: Case data shall not be shared with external parties without authorization.

DC-4: Firm data shall remain on local server (not sent to cloud services except embeddings for Pinecone).

DC-5: Chat logs shall not contain confidential client information.

DC-6: Users shall have right to access their own data.

DC-7: Data export functionality shall be available for compliance.

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8. Appendices

8.1 Appendix A: Glossary (Expanded)

Term	Full Form	Definition
Accused	-	Person charged with criminal offense
Acquittal	-	Judgment that a person is not guilty
API	Application Programming Interface	Set of protocols for building software
Bail	-	Temporary release of accused pending trial
BERT	Bidirectional Encoder Representations from Transformers	ML model for natural language understanding
BNS	Bharatiya Nyaya Sanhita	New Indian Penal Code (replaced IPC)
Conviction	-	Judgment that a person is guilty

CORS	Cross-Origin Resource Sharing	Mechanism allowing restricted resources on web page
CrPC	Code of Criminal Procedure	Procedural law for criminal cases in India
CSRF	Cross-Site Request Forgery	Type of web security vulnerability
Django	-	Python web framework
Embedding	-	Vector representation of text
FIR	First Information Report	Initial police report of cognizable offense
GridFS	-	MongoDB specification for storing large files
IPC	Indian Penal Code	Criminal code of India (now BNS)
JWT	JSON Web Token	Compact means of transmitting information
LLM	Large Language Model	AI model trained on large text corpus
NER	Named Entity Recognition	NLP task of identifying entities in text
NLP	Natural Language Processing	AI field concerned with language understanding
OCR	Optical Character Recognition	Converting images to text
ORM	Object-Relational Mapping	Programming technique for database access
Petitioner	-	Person who files petition in court
Plaint	-	Written statement of plaintiff's claim
PostgreSQL	-	Open-source relational database
RAG	Retrieval Augmented Generation	AI technique combining retrieval and generation
RBAC	Role-Based Access Control	Access control based on user roles
Respondent	-	Person against whom petition is filed
REST	Representational State Transfer	Architectural style for APIs
SARFAESI	Securitisation and Reconstruction of Financial Assets and Enforcement of Security Interest Act	Act for recovery of defaulted loans
Sentence	-	Punishment imposed by court

SHAP	SHapley Additive exPlanations	Method for explaining ML predictions
SSL/TLS	Secure Sockets Layer / Transport Layer Security	Cryptographic protocols for secure communication
Vector Database	-	Database optimized for vector similarity search
WCAG	Web Content Accessibility Guidelines	Accessibility standards

8.2 Appendix B: Use Case Diagrams

Note: Detailed use case diagrams shall be created during design phase. Key use cases include:

User Management

- Login
- Create User Account
- Reset Password
- Manage Permissions

Case Management

- Upload Case Document
- View Case Details
- Edit Case Metadata
- Delete Case

AI Features

- Generate Prediction
- Find Similar Cases
- Ask Chatbot Question
- Generate Court Script
- Explain Prediction

Repository

- Search Cases
- Filter Cases
- Export Report
- Create Case Collection

Administration

- View Audit Logs
- Configure System

- Manage Backups
- View Statistics

Networking Platform

- Create Profile
- Post Job
- Apply to Job
- Message Applicant

8.3 Appendix C: Data Flow Diagrams

High-Level Data Flow:

```
User → Upload Document → System
System → Extract Text → PostgreSQL (metadata)
System → Store File → MongoDB (full document)
System → Generate Embedding → Pinecone (vectors)
System → Run NER → PostgreSQL (entities)
System → Display → User

User → Request Prediction → System
System → Load Features → PostgreSQL
System → Run ML Model → Prediction Result
System → Generate Explanation → SHAP
System → Store Result → PostgreSQL
System → Display → User

User → Search Query → System
System → Generate Query Embedding → Vector
System → Search → Pinecone
Pinecone → Similar Cases → System
System → Display → User

User → Ask Chatbot → System
System → Search Context → Pinecone
System → Build Prompt → LLM (GPT-4)
LLM → Answer → System
System → Display → User

User → Generate Script → System
System → Retrieve Cases → PostgreSQL + Pinecone
System → Format Script → Script Generator
System → Return Document → User
```

8.4 Appendix D: Entity-Relationship Diagrams

Key Entities and Relationships:

```
USER (user_id, name, email, password_hash, role)
  |
  | 1:N (uploaded_by)
  |
CASE (case_id, case_number, type, court, date, sections, outcome)
  |
```

```
  | 1:N
  |
PREDICTION (prediction_id, case_id, model_type, value,
explanation)

USER
  |
  | 1:N (performed_by)
  |
AUDIT_LOG (log_id, user_id, action, case_id, timestamp)

CASE
  |
  | 1:1
  |
DOCUMENT (doc_id, case_id, full_text, entities, metadata)

LAWYER_PROFILE (profile_id, user_id, experience, specialization)
  |
  | 1:N (posted_by)
  |
JOB_POSTING (job_id, title, description, domain, location)
  |
  | N:M (via APPLICATION)
  |
STUDENT_PROFILE (profile_id, user_id, education, interests)

APPLICATION (app_id, job_id, student_id, status, date)
```

8.5 Appendix E: System Limitations

L-1: The system does not provide legally binding advice and all outputs must be verified by qualified lawyers.

L-2: AI predictions are based on historical data and may not account for recent legal changes or unique case circumstances.

L-3: The system focuses on Supreme Court judgments; High Court and District Court cases are not included in initial dataset.

L-4: OCR accuracy depends on scan quality; poor quality scans may result in errors.

L-5: The system requires internet connectivity for cloud services (Pinecone, OpenAI API).

L-6: Non-English legal documents are not supported.

L-7: The system cannot handle extremely large files (> 25 MB) directly.

L-8: Real-time collaboration features are not included in initial version.

L-9: Mobile apps are not included; web interface is responsive but optimized for desktop use.

L-10: The system does not integrate with existing court filing systems.

8.6 Appendix F: Future Enhancements

FE-1: Integration with High Court and District Court databases

FE-2: Support for regional language judgments

FE-3: Mobile applications (iOS and Android)

FE-4: Real-time collaboration features

FE-5: Integration with court filing systems (e-filing)

FE-6: Advanced analytics and reporting

FE-7: Client portal for case tracking

FE-8: Automated legal notice generation

FE-9: Voice-based interaction with chatbot

FE-10: Multi-tenant deployment for multiple law firms

FE-11: Integration with legal research databases (Manupatra, SCC Online)

FE-12: Blockchain-based audit trail for immutability

8.7 Appendix G: Assumptions Log

Assumption ID: A-001

- **Assumption:** Kaggle dataset will remain accessible throughout development
- **Impact if False:** HIGH - Need alternative data source
- **Mitigation:** Download and archive dataset locally

Assumption ID: A-002

- **Assumption:** Law firm will provide dedicated server with specified hardware
- **Impact if False:** MEDIUM - May need to downgrade features or use cloud
- **Mitigation:** Cloud deployment option as backup

Assumption ID: A-003

- **Assumption:** Users have basic computer literacy and can navigate web applications
- **Impact if False:** LOW - May need more comprehensive training
- **Mitigation:** Create detailed user guides and video tutorials

Assumption ID: A-004

- **Assumption:** OpenAI API will be available and pricing will remain reasonable
- **Impact if False:** MEDIUM - Switch to self-hosted Llama 3

- **Mitigation:** Design system to support multiple LLM backends

Assumption ID: A-005

- **Assumption:** Legal regulations will allow use of AI for decision support
- **Impact if False:** HIGH - System cannot be deployed
- **Mitigation:** Clear disclaimers, lawyer verification requirements

Assumption ID: A-006

- **Assumption:** The Software Requirements Specification (SRS) and system development are subject to project time constraints; system modules and features will be implemented based on available time.
- **Impact if False: MEDIUM to HIGH** – Delays may prevent full implementation of planned modules and features.
- **Mitigation:**
  - Prioritize core and critical system modules
  - Use a phased or incremental development approach
  - Defer or drop non-essential features if time constraints do not permit
  - Maintain clear documentation of excluded or postponed modules

8.8 Appendix H: Risk Register

Risk ID	Risk Description	Probability	Impact	Mitigation Strategy
R-001	Data source becomes unavailable	Low	High	Archive dataset locally
R-002	ML models underperform (< 75% accuracy)	Medium	High	Iterate on feature engineering, collect more data
R-003	External API costs exceed budget	Medium	Medium	Implement usage limits, consider self-hosted alternatives
R-004	Server hardware failure	Low	High	Implement daily backups, have spare hardware
R-005	Security breach / data leak	Low	Critical	Implement comprehensive security measures,

				regular audits
R-006	User adoption is low	Medium	Medium	Comprehensive training, user-friendly design
R-007	Performance degradation with scale	Medium	Medium	Implement caching, database optimization
R-008	Regulatory compliance issues	Low	Critical	Legal review before deployment, clear disclaimers
R-009	Development timeline overruns	Medium	Medium	Regular milestone reviews, scope management
R-010	Team member unavailability	Medium	Medium	Cross-training, documentation

8.9 Appendix I: Approval Signatures

Document Prepared By:

Development Team:

- Muneeb Sawez - Backend Lead
- Mohammed Fawaz Mulla - Frontend Lead
- Sayyad Abrar Ali - Data Processing Lead
- Kaushik Rajesh Parvate - ML/Security Lead

Document Reviewed By:

Client Representative:

- Name: _____
- Title: _____
- Date: _____
- Signature: _____

Project guide :

- Name: _____
- Title: _____
- Date: _____
- Signature: _____

END OF DOCUMENT

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This SRS document is a living document and will be updated as requirements evolve during the development process. All changes must be approved by stakeholders and documented in the version history.